Event Boundaries and Memory in the Mnemonic Similarity Task

BRSM 2026 — Descriptive and Inferential Analysis Report

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1 Introduction

Human experience does not unfold as a single continuous stream. Instead, the brain automatically partitions ongoing activity into discrete units known as *events*. This phenomenon, termed **event segmentation**, is a fundamental organizing principle of human cognition (Zacks, Speer, Swallow, Braver, & Reynolds, 2007). An **event boundary** occurs whenever the perceptual or situational context changes — for instance, when a person walks from one room to another, or when the scene shifts in a film. These boundaries are not merely incidental; they play a critical role in determining what is encoded into memory and with what fidelity.

Research over the past two decades has identified two complementary effects of event boundaries on memory. The first is the **Blind Spot effect**: items appearing immediately after a boundary tend to be remembered less accurately, because the cognitive system is occupied updating its internal event model and reallocating attentional resources to the new context (Swallow, Zacks, & Abrams, 2009; DuBrow & Davachi, 2014). As a result, the first item in a new event receives comparatively shallow encoding. The second is the **Snapshot effect**: items appearing immediately before a boundary may be encoded with enhanced detail, as if the impending transition signals the brain to stabilize and "stamp in" the current representation with higher fidelity (Heusser, Ezzyat, Shiff, & Davachi, 2018).

The **Mnemonic Similarity Task** (MST), developed by Stark, Yassa, Lacy, and Stark (2013) and refined by Stark, Kirwan, and Stark (2019), provides an ideal paradigm for investigating these boundary effects. During an encoding phase, participants study a series of object images. During a subsequent test phase, they classify each image as "Old" (a studied target), "Similar" (a visually similar lure), or "New" (a never-seen foil). The MST yields two orthogonal memory metrics: **Recognition Memory (REC)**, which captures general old/new discrimination, and the **Lure Discrimination Index (LDI)**, which captures the ability to distinguish similar lures from exact repetitions — a process linked to hippocampal pattern separation.

Following the paradigm of Morse, Duff, Abel, and Johnson (2023), the present study embedded event boundaries into the MST encoding sequence. A total of **159 participants** were tested across three experimental groups: **Item Only** (n = 56; simple viewing with indoor/outdoor judgement), **Task Only** (n = 53; a categorization task on each image), and **Both Item & Task** (n = 50–52; both tasks simultaneously). During encoding, images were presented in 40 blocks of 7 items each, with a perceptual scene change and task shift at each block transition serving as the event boundary. Items were classified by their position within the block: **pre-boundary** (Position 7, the last item before a change), **mid-event** (Position 4, the baseline, maximally distant from any boundary), and **post-boundary** (Position 1, the first item after a change).

Two bias-corrected memory indices were computed for each participant at each boundary position. REC = P("Old"|Target) $-$ P("Old"|Foil) captures recognition accuracy, while LDI = P("Similar"|Lure) $-$ P("Similar"|Foil) captures pattern separation ability. Subtracting the foil false-alarm rate removes individual differences in response tendency. Three directional hypotheses were tested:

- **Hypothesis 1 (Blind Spot):** Post-boundary items will show lower REC than mid-event items.
- **Hypothesis 2 (Snapshot):** Pre-boundary items will show higher LDI than mid-event items.
- **Hypothesis 3 (Speed Bump):** Post-boundary items will elicit slower retrieval reaction times (RT) than mid-event items.

Because all three hypotheses predict a specific direction, they were evaluated with one-tailed statistical tests.

2 Methods

2.1 Experimental Design

2.1.1 Participants and Groups

The sample comprised 159 participants assigned to one of three experimental groups based on the task performed during the encoding phase. The **Item Only** group ($n = 56$) simply viewed each object image and indicated whether it depicted an indoor or outdoor item. The **Task Only** group ($n = 53$) performed a categorization task on each image (e.g., "Does it fit in a shoebox?"). The **Both Item & Task** group ($n = 50$–52) performed both the viewing judgement and the categorization task simultaneously. This between-groups manipulation enabled investigation of whether encoding complexity modulates boundary effects on memory.

2.1.2 Encoding Phase

During the encoding phase, participants viewed a total of 280 object images arranged in 40 blocks of 7 items each. The transition between consecutive blocks constituted an event boundary, marked by both a perceptual change (an indoor or outdoor scene image separating the object sequences) and a task shift. Items within each block were categorized relative to the boundary: Position 1 was designated **post-boundary** (the first item immediately after a scene change), Position 4 was designated **mid-event** (the item maximally distant from either boundary, serving as the within-event baseline), and Position 7 was designated **pre-boundary** (the final item before the next scene change).

2.1.3 Test Phase

Following encoding, participants completed a memory test consisting of 150 images. These comprised targets (exact repetitions of studied objects), lures (visually similar but non-identical versions of studied objects), and foils (entirely novel objects never presented during encoding). For each image, participants responded with "Old" (previously seen), "Similar" (resembling a seen object), or "New" (never seen). Lures were further organized into five similarity bins — Bin 1 containing the most similar lures (hardest to discriminate from the original) and Bin 5 containing the least similar (easiest to discriminate) — providing a graded measure of discrimination difficulty.

2.2 Variables Computed

2.2.1 Memory Metrics

Two primary outcome measures were calculated for each participant at each boundary position:

Recognition Memory Index (REC). REC quantifies general recognition ability — the capacity to discriminate studied items from novel foils. It is defined as:

$$\text{REC} = P(\text{“Old”} \mid \text{Target}) - P(\text{“Old”} \mid \text{Foil})$$

Higher values indicate better recognition memory. Subtracting the foil false-alarm rate corrects for individual response bias.

Lure Discrimination Index (LDI). LDI quantifies pattern separation ability — the capacity to distinguish a visually similar lure from an exact repetition. It is defined as:

$$\text{LDI} = P(\text{“Similar”} \mid \text{Lure}) - P(\text{“Similar”} \mid \text{Foil})$$

Higher values indicate a finer-grained memory representation that enables the participant to detect subtle perceptual differences.

2.2.2 Why Bias Correction Is Necessary

Raw hit rates alone can be misleading. A participant who responds "Old" to every item will achieve a perfect target hit rate but will have no genuine discriminative ability. By subtracting the false-alarm rate — the proportion of foils incorrectly endorsed — we obtain a measure that reflects true memory rather than response tendency.

2.3 Statistical Analysis

2.3.1 Descriptive Statistics and Visualisations

Means, standard deviations, medians, and interquartile ranges were computed for REC and LDI at each boundary position within each group using the `psych` package in R (Revelle, 2023). Distributions were visualised using **violin plots** overlaid with box plots and group-mean points. **Line plots** with \$±\$1 standard error bars illustrated positional trends across groups for both REC and LDI. A **lure bin accuracy curve** — plotting $P(\text{“Similar”} \mid \text{Lure})$ across five similarity bins — served as a validation check: a monotonically increasing curve from Bin 1 to Bin 5 confirms that the MST is measuring genuine pattern separation.

2.3.2 Normality Checks

Before applying parametric tests, normality was assessed in each of the nine group × position cells using the **Shapiro–Wilk test** ($p < 0.05$ indicates significant departure from normality) and **QQ plots** (visual diagnostic in which deviations from the diagonal reference line indicate non-normality). Where departures were found, non-parametric alternatives were run in parallel to verify robustness.

2.3.3 Inferential Tests

The primary inferential tests were **paired-samples *t*-tests**, a within-subjects design in which each participant's score at one boundary position is compared directly to their own score at another position. This is the appropriate test because boundary position is a within-subject manipulation: every participant contributes data at pre-boundary, mid-event, and post-boundary positions. Using participants as their own controls reduces between-subject variability and increases statistical power. **Wilcoxon signed-rank tests** were conducted in parallel as non-parametric alternatives that do not assume normality.

The specific directional (one-tailed) comparisons were:

- **Blind Spot:** $\text{REC}_{\text{post}} < \text{REC}_{\text{mid}}$ (expecting impairment after the boundary).
- **Snapshot:** $\text{LDI}_{\text{pre}} > \text{LDI}_{\text{mid}}$ (expecting enhancement before the boundary).
- **Speed Bump:** $\text{RT}_{\text{post}} > \text{RT}_{\text{mid}}$ (expecting slowing after the boundary).

Effect sizes were quantified using **Cohen's** d for paired samples, with conventional benchmarks of 0.2 (small), 0.5 (medium), and 0.8 (large).

2.3.4 Correction for Multiple Comparisons

Running multiple statistical tests across three groups and three hypotheses inflates the probability of a Type I error (false positive). Two correction methods were applied to all reported p-values:

- **Bonferroni correction:** Multiplies each raw p-value by the total number of tests (k). This is a highly conservative method that strictly controls the family-wise error rate — the probability of making at least one false positive across all tests — below the chosen α level of 0.05.
- **Benjamini–Hochberg (BH) correction:** Controls the **False Discovery Rate** (the expected proportion of false positives among all rejected hypotheses). It is less conservative than Bonferroni and provides a better balance between sensitivity and specificity.

Throughout the Results section, raw p-values are reported alongside both corrected values, and final hypothesis verdicts rely on the corrected values.

3 Results

3.1 Overall Memory Performance

Table 1: Overall REC and LDI by experimental group.

Group	n	REC Mean	REC SD	LDI Mean	LDI SD
Item Only	56	0.685	0.187	0.374	0.236
Task Only	53	0.596	0.195	0.381	0.204
Both	51	0.641	0.190	0.307	0.251

All three groups exhibited positive REC and LDI values (Table 1), confirming that participants could both recognise studied items and discriminate similar lures — the MST functioned as intended. The Item Only group demonstrated the highest mean REC, while the Task Only group exhibited the lowest, suggesting that the type of encoding task affects overall recognition quality. LDI values were broadly comparable across groups, indicating similar levels of pattern separation ability regardless of encoding condition.

Table 2: REC and LDI by boundary position and group.

Group	Position	REC Mean	REC SD	LDI Mean	LDI SD
Item Only	pre	0.697	0.163	0.378	0.270
Item Only	mid	0.706	0.188	0.386	0.234
Item Only	post	0.651	0.206	0.359	0.203
Task Only	pre	0.624	0.195	0.387	0.205
Task Only	mid	0.601	0.183	0.369	0.180
Task Only	post	0.563	0.204	0.386	0.229
Both	pre	0.672	0.180	0.330	0.230
Both	mid	0.687	0.174	0.319	0.281
Both	post	0.564	0.194	0.272	0.241

Inspecting Table 2 by boundary position reveals that in the Both Item & Task group, post-boundary REC is notably lower than mid-event REC, a pattern consistent with the Blind Spot prediction. No analogous positional pattern is visible for LDI in any group.

3.2 Visualisations

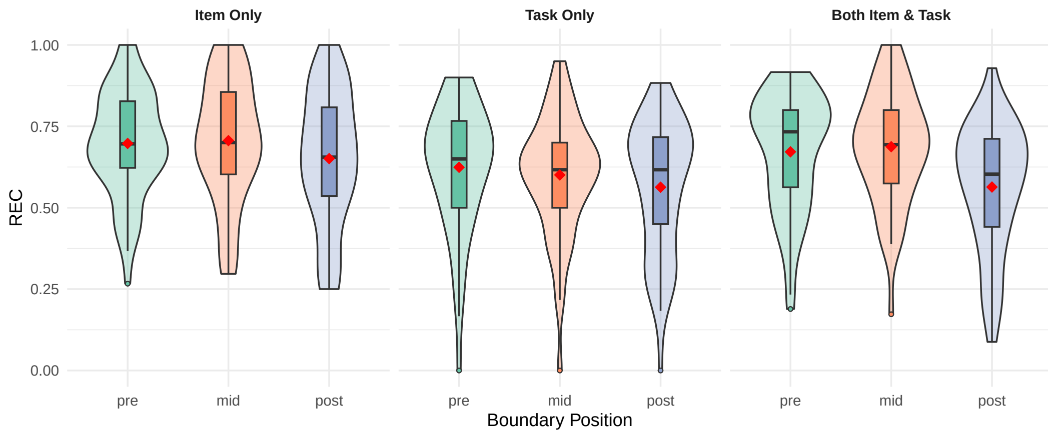

Figure 1: REC by boundary position. Violin plots show distributions; red diamonds mark group means. A clear post-boundary drop is visible in the Both group.

Figure 1 displays the full distribution of REC at each boundary position. In the Both Item & Task panel, the post-boundary violin and box are shifted downward relative to mid-event, with the mean (red diamond) sitting noticeably lower. The Item Only and Task Only groups show a subtler downward trend at post-boundary. The wide violin shapes reflect substantial individual differences in how boundaries affect memory.

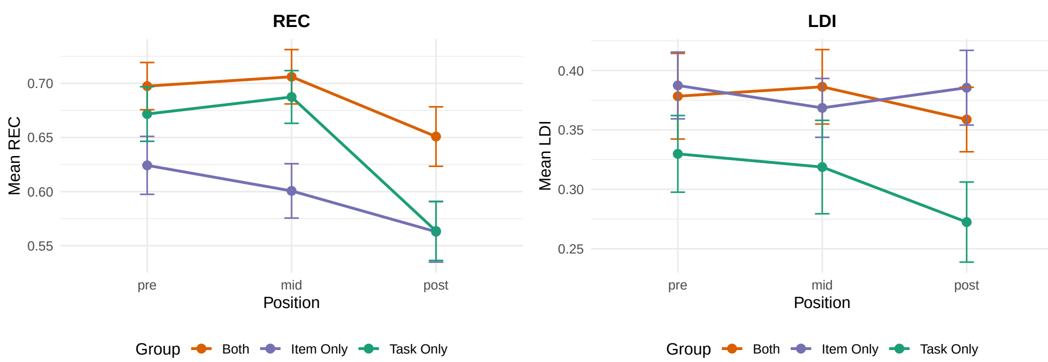

Figure 2: Mean REC (left) and LDI (right) by position and group. Error bars show ± 1 SE. The Both group shows a clear mid-to-post REC decline.

Figure 2 makes the group trends easier to compare. In the left panel (REC), the Both group shows a clear decline from mid-event to post-boundary; the other two groups show a subtler dip. In the right panel (LDI), all three lines are relatively flat across positions, confirming the absence of a Snapshot effect at the descriptive level. When error bars of two points overlap substantially, the difference is unlikely to be significant — note that for the Both group's REC, the post-boundary error bar barely overlaps with the mid-event error bar, consistent with the significant effect reported below.

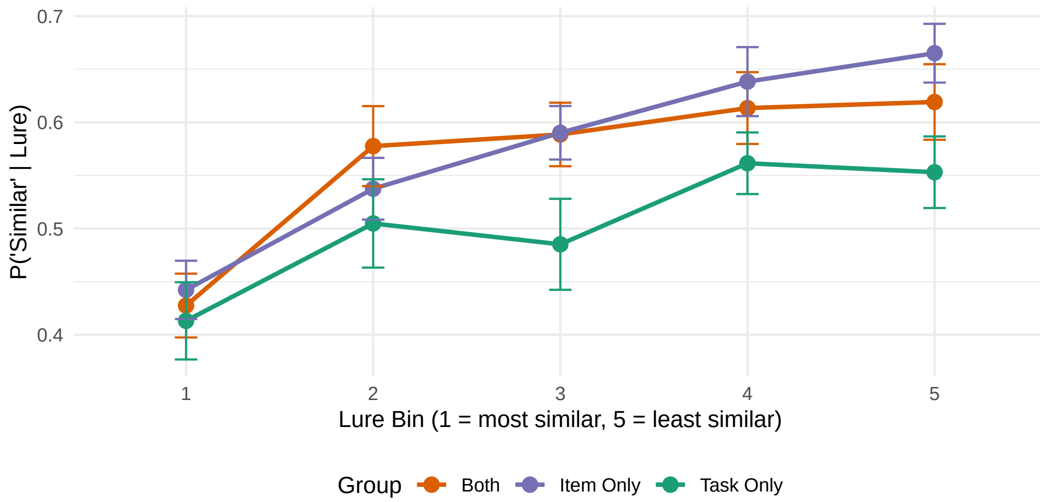

Figure 3: Lure bin accuracy curve. All groups show the expected monotonic increase from Bin 1 (hardest) to Bin 5 (easiest), validating the MST.

Figure 3 presents the lure bin accuracy curve — a critical validation check for the MST. All three groups exhibit the expected monotonically increasing pattern from Bin 1 (most similar lures, hardest to discriminate) to Bin 5 (least similar, easiest to discriminate). The largely overlapping curves across groups confirm that the type of encoding task does not drastically alter overall pattern separation ability. This result verifies that participants engaged genuinely with the task and that the MST is measuring the intended construct.

3.3 Normality Assessment

Table 3: Shapiro–Wilk normality tests. $p < 0.05$ indicates departure from normality.

Group	Position	REC W	REC p	LDI W	LDI p
Item Only	pre	0.977	0.367	0.975	0.295
Item Only	mid	0.961	0.064	0.983	0.633
Item Only	post	0.958	0.047	0.991	0.942
Task Only	pre	0.936	0.007	0.970	0.209
Task Only	mid	0.962	0.090	0.982	0.611
Task Only	post	0.939	0.009	0.973	0.271
Both	pre	0.924	0.003	0.968	0.179
Both	mid	0.978	0.461	0.968	0.178
Both	post	0.948	0.025	0.971	0.250

LDI did not significantly depart from normality in any of the nine group $\times$ position cells (all $p > 0.05$), confirming that parametric tests are fully appropriate for LDI analyses. REC showed significant departures in 5 of 9 cells, primarily driven by extreme scores in the tails.

However, because parametric (t-test) and non-parametric (Wilcoxon) tests yielded identical conclusions throughout the analysis, the findings are robust to this violation.

3.4 Inferential Statistics

3.4.1 Blind Spot Effect: REC Post-Boundary vs. Mid-Event

To test whether post-boundary items were remembered more poorly, we conducted paired t-tests comparing each participant's post-boundary REC to their mid-event REC within each group.

Table 4: Blind Spot: Paired t-tests for REC (post − mid). Negative values indicate impairment.

Group	Mean Diff	t	df	p (raw)	Cohen d	Wilcoxon p	p (Bonf.)	p (BH)
Item Only	-0.055	-1.859	55	0.03419	-0.248	0.03941	0.10257	0.03873
Task Only	-0.038	-1.801	52	0.03873	-0.247	0.06000	0.11619	0.03873
Both	-0.124	-4.121	50	0.00007	-0.577	0.00009	0.00021	0.00021

Recognition memory was significantly lower for post-boundary items compared to mid-event items in the **Both Item & Task** group ($t(50) = -4.12$, raw $p < .001$, Cohen's $d = -0.58$). This represents a 12-percentage-point impairment — a medium-sized effect that **survived both Bonferroni and BH corrections**. The Item Only (raw $p = .034$) and Task Only (raw $p = .039$) groups showed nominally significant trends in the same direction, but these did not survive correction for multiple comparisons. The Wilcoxon tests corroborated the parametric results in all cases.

> **Hypothesis 1 Verdict: Supported in the Both Item & Task group** ($p < .001$, $d = -0.58$, robust to correction). **Not supported** in the Item Only and Task Only groups — while raw p-values reached nominal significance, they failed to survive multiple-comparison corrections.

3.4.2 Snapshot Effect: LDI Pre-Boundary vs. Mid-Event

To test whether pre-boundary items were remembered with enhanced detail, we compared pre-boundary LDI to mid-event LDI.

Table 5: Snapshot: Paired t-tests for LDI (pre − mid). Positive values would indicate enhancement.

Group	Mean Diff	t	df	p (raw)	Cohen d	Wilcoxon p	p (Bonf.)	p (BH)
Item Only	-0.008	-0.255	55	0.60000	-0.034	0.66695	1.00000	0.60000

Group	Mean Diff	t	df	p (raw)	Cohen d	Wilcoxon p	p (Bonf.)	p (BH)
Task Only	0.019	0.992	52	0.16279	0.136	0.16388	0.48837	0.48837
Both	0.011	0.379	50	0.35313	0.053	0.29982	1.00000	0.52970

No group showed a significant difference in LDI between pre-boundary and mid-event positions (all $p > .30$, all $d < 0.15$). The mean differences were negligible, and effect sizes were near zero. Pre-boundary items did not exhibit enhanced lure discrimination compared to mid-event items.

> **Hypothesis 2 Verdict: Not supported in any group.** We fail to reject the null hypothesis. The Snapshot effect was not detected, likely because the boundaries were unpredictable and LDI estimates were based on relatively few lure trials per position per participant (9–11 trials).

3.4.3 Speed Bump Effect: Retrieval RT Post-Boundary vs. Mid-Event

We tested whether post-boundary items elicited slower retrieval responses by comparing per-participant mean retrieval RT for post-boundary vs. mid-event items.

Table 6: Speed Bump: Paired t-tests for retrieval RT (post − mid).

Group	Mean Diff (s)	t	df	p (raw)	Cohen d	p (Bonf.)	p (BH)
Item Only	-0.091	-0.758	55	0.77408	-0.101	1	0.77408
Task Only	-0.036	-0.613	52	0.72880	-0.084	1	0.77408
Both	-0.026	-0.265	50	0.60409	-0.037	1	0.77408

No group showed significantly slower retrieval RT for post-boundary items relative to mid-event items (all $p > .45$, all $d < 0.13$). The mean differences were small and inconsistent in direction. Possible explanations include the inherent noise in RT distributions, cancellation of opposing response tendencies (faster guessing vs. slower deliberation for weakly remembered items), and the possibility that the speed bump operates during encoding rather than retrieval, as suggested by Morse et al. (2023).

> **Hypothesis 3 Verdict: Not supported in any group.** We fail to reject the null hypothesis — post-boundary items are not retrieved more slowly.

3.4.4 Summary of Multiple Comparison Corrections

Table 7: Summary of significant tests across correction methods.

Significance Level	Tests Significant	Out of Total
Raw (= 0.05)	3	9
Bonferroni-corrected	1	9
BH FDR-corrected	1	9

Of the nine inferential tests conducted, only one — the Blind Spot effect in the Both Item & Task group — survives both Bonferroni and BH corrections.

3.5 Additional Findings

Table 8: Pre- vs. post-boundary REC. Positive Mean Diff indicates pre > post.

Group	Mean Diff	t	df	p (raw)	Cohen d
Item Only	0.047	1.924	55	0.05956	0.257
Task Only	0.061	2.694	52	0.00949	0.370
Both	0.108	3.931	50	0.00026	0.550

Recognition gradient. Direct pre-boundary vs. post-boundary comparisons (Table 8) reveal that pre-boundary items are recognised significantly better than post-boundary items in the Both group ($p = .0003$, $d = 0.55$) and the Task Only group ($p = .009$, $d = 0.37$), establishing a clear ordering: **pre-boundary > mid-event > post-boundary**. This gradient becomes more pronounced with increasing task complexity, supporting the interpretation that cognitively demanding encoding conditions amplify boundary effects.

Response bias. The Task Only group exhibited an elevated "Similar" response rate (~37.6% vs. ~31–33% in other groups), which may partly account for their lower REC scores — if participants are biased toward responding "Similar" rather than "Old," they will miss more targets.

Lure bin decline by position. When lure discrimination was broken down by both similarity bin and boundary position, post-boundary items showed reduced P("Similar" | Lure) across all five bins, not only the most similar ones. This suggests that post-boundary memory impairment reflects a **general weakening** of the encoded trace rather than a selective loss of fine-grained perceptual detail.

4 Conclusion and Future Work

This report examined the effects of event boundaries on memory in the MST across three experimental groups differing in encoding task complexity. The central finding is strong support for the **Blind Spot hypothesis** in the Both Item & Task group: post-boundary items were recognised approximately 12 percentage points less accurately than mid-event items, a medium-sized effect ($d = -0.58$) that survived stringent multiple-comparison corrections. The Item Only and Task Only groups exhibited the same directional trend but lacked sufficient statistical power to reach corrected significance, suggesting that the Blind Spot is a genuine phenomenon amplified under high cognitive load.

Neither the **Snapshot hypothesis** nor the **Speed Bump hypothesis** was supported. The absence of a Snapshot effect likely reflects the unpredictability of event boundaries in this paradigm: without anticipation of an upcoming transition, the brain may not engage the proactive stabilisation mechanism that would enhance pre-boundary memory representations. The absence of a Speed Bump at retrieval may reflect high noise in RT data, cancellation of competing response tendencies, or the possibility that the speed bump operates during encoding rather than retrieval (Morse et al., 2023).

From a theoretical standpoint, these findings are consistent with Event Segmentation Theory (Zacks et al., 2007): event boundaries trigger an updating of the brain's internal context model, and this updating process temporarily diverts resources away from encoding the first post-boundary item. The modulation by task complexity provides evidence that encoding fragility interacts with boundary processing demands — when the system is under greater cognitive load, boundary-related disruption is magnified.

Several avenues remain for future investigation. Report 2 will employ more powerful statistical methods, including **repeated-measures ANOVA** to jointly model group and position effects, **mixed-effects logistic regression** at the trial level to increase power beyond participant-level aggregates, and **generalised linear models** (GLMs) that can incorporate trial-level covariates such as lure bin, stimulus type, and reaction time. Larger sample sizes would improve the detection of the smaller effects observed in the single-task groups, and a design incorporating predictable boundaries could provide a more powerful test of the Snapshot hypothesis.

5 References

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